# Chapter 1

# Black-Scholes Model

## 1.1 Model Setup: Stock and Bond Dynamics

Call option price formula is shown by

$$C = S_0\mathbb{N}(d_1) - Ke^{-rT}\mathbb{N}(d_2)$$

Put option price can be calculated by Put-Call Parity

$$P + S_0 = C + Ke^{-rT}$$

Assuming we have a portfolio with underlying stock $S$ and risk-free asset $B$, differential equations are shown by

$$dB_t = rB_t\mathrm{d}t \tag{1.1}$$

$$dS_t = \mu S_t\mathrm{d}t + \sigma S_t\mathrm{d}W_t \tag{1.2}$$

where $\mathrm{d}W_t = \varepsilon\sqrt{\mathrm{d}t}$; $\varepsilon \sim \mathcal{N}(0,1)$, which is a Ito process. $\mu$ is drift (deterministic) and $\mathrm{d}W_t$ is a diffusion (uncertainty).

### Solving $S_t$ and $B_t$ from differential equations

$B_t$ and $S_t$ can be expressed as

$$B_t = B_0e^{rt} \tag{1.3}$$

***Proof:***

$$\begin{aligned}
\int \frac{dB_t}{B_t} &= \int r dt \\
\int_0^t \ln B_t &= \int_0^t r dt \\
\ln(B_t/B_0) &= r * t \\
B_t &= B_0e^{rt}
\end{aligned}$$

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$$S_t = S_0\exp\left[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right] \tag{1.4}$$

***Proof:*** Let $Y_t = \ln S_t$

$$\begin{aligned}
\mathrm{d}Y_t &= \frac{\partial Y_t}{\partial t}dt + \frac{\partial Y_t}{\partial S_t}\mathrm{d}S_t + \frac{1}{2}\frac{\partial^2 Y_t}{\partial S_t^2}\mathrm{d}S_t^2 \\
&= \frac{1}{S_t}(\mu S_t\mathrm{d}t + \sigma S_t\mathrm{d}W_t) + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)\sigma^2S_t^2\mathrm{d}t \\
&= (\mu - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma\mathrm{d}W_t
\end{aligned}$$

we get

$$\begin{aligned}
Y_t &= Y_0 + \int_0^t (\mu - \frac{1}{2}\sigma^2)\mathrm{d}s + \int_0^t \sigma \mathrm{d}W_s \\
&= Y_0 + (\mu - \frac{1}{2})t + \sigma W_t
\end{aligned}$$

Given $Y_t = \ln S_t$ Finally, we have to show

$$\begin{aligned}
\ln S_t &= \ln S_0 + (\mu - \frac{1}{2})t + \sigma W_t \\
\implies S_t &= S_0 \exp\left[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right]
\end{aligned}$$

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## 1.2 Risk-Neutral Martingale Pricing Derivation

**Proposition 1.2.1**

In the Black-Scholes model, the price process $\Pi_t$ for every traded asset, be it the underlying or derivative asset, has the property that the normalised price process

$$Z_t = \frac{\Pi_t}{B_t}$$

is a martingale under the $\mathbb{Q}$ measure.

In this case,

$$Z_t = \frac{S_t}{B_t}$$

Its differential equation can be written as

$$dZ_t = d\left(\frac{S_t}{B_t}\right)$$

$d\left(\frac{1}{B_t}\right)$ can be calculated by Ito Lamma

$$\begin{aligned}
d\left(\frac{1}{B_t}\right) &= 0dt + \left(-\frac{1}{B_t^2}\right)dB_t + \frac{1}{2}\cdot\left(-\frac{2}{B_t^3}\right)(dB_t)^2 \\
&= -\frac{1}{B_t^2}\cdot rB_t dt - 0 \\
&= -\frac{1}{B_t} r dt
\end{aligned}$$

Using product rule, then we have

$$\begin{aligned}
d\left(\frac{S_t}{B_t}\right) &= \frac{1}{B_t}\cdot dS_t + S_t \cdot \mathrm{d}\left(\frac{1}{B_t}\right) + dS_t \cdot \mathrm{d}\left(\frac{1}{B_t}\right) \\
&= \frac{1}{B_t}\cdot(\mu S_t dt + \sigma S_t dW_t) + S_t \cdot \left(-\frac{r}{B_t}\right)dt + 0 \\
&= \frac{S_t}{B_t}\mu dt + \frac{S_t}{B_t}\sigma dW_t - \frac{S_t}{B_t} r dt \\
&= \frac{S_t}{B_t}\left(\mu - r\right)dt + \frac{S_t}{B_t}\sigma dW_t
\end{aligned}$$

Namely under $\mathbb{P}$ measure

$$dZ_t = Z_t(\mu - r)dt + Z_t \sigma dW_t$$

### Transfer $\mathbb{P}$ to $\mathbb{Q}$ measure

**Theorem 1.2.1** Novikov Condition

$\theta$ satisfies the Novikov Condition if $\mathbb{E}\left[\exp\left(\frac{1}{2}\int_0^T \theta_s^2 \mathrm{d}s\right)\right] < \infty$

If $\theta$ satisfies Novikov condition, the process $\mathcal{M}^\theta$ defined as

$$\mathcal{M}_t^\theta = \exp\left(\int_0^t \theta_s \mathrm{d}X_s - \frac{1}{2}\int_0^t \theta_s^t \mathrm{d}s\right), t \in [0, T]$$

is a martingale.

**Theorem 1.2.2** Girsanovs Theorem

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \exp\left(\int_0^t \theta_s \mathrm{d}X_s - \frac{1}{2}\int_0^t \theta_s^2 \mathrm{d}s\right)$$

In this case, the process $X^{\mathbb{Q}}$ is defined as

$$X_t^{\mathbb{Q}} = X_t^{\mathbb{P}} - \int_0^t \theta_s \mathrm{d}s, t \in [0, T]$$

is a standard Brownian Motion.

From Girsanovs Theorem, we have

$$W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \theta_s \mathrm{d}s$$

and

$$\mathrm{d}W_t^{\mathbb{Q}} = \mathrm{d}W_t + \theta_t \mathrm{d}t$$

$$\begin{aligned} dZ_t &= (\mu - r)Z_t dt + \sigma Z_t dW_t^{\mathbb{P}} \\ &= (\mu - r)Z_t dt + \sigma Z_t(\theta_t dt + dW_t^{\mathbb{Q}}) \\ &= (\mu - r + \sigma\theta_t)Z_t dt + \sigma Z_t dW_t^{\mathbb{Q}} \end{aligned}$$

$dZ_t$ is a martingale under $\mathbb{Q}$ dynamics Therefore

$$\mu - r + \sigma\theta_t = 0$$

$\theta_t$ can be expressed as

$$\theta_t = \frac{\mu - r}{\sigma}$$

And $\theta_t$ is the Sharpe Ratio. $\mu$ is the return of underlying stock $S$ and $r$ is the risk-free rate.

Under $\mathbb{Q}$ measure

$$dZ_t = \sigma Z_t dW_t^{\mathbb{Q}}$$

## Portfolio Replication

Return back to our portfolio. $\Phi_t^S$ and $\Phi_t^B$ are the weights of the stock and risk-free asset repectively.

The value of the portfolio $V_t$ and its differential equation are shown as

$$V_t = \Phi_t^S S_t + \Phi_t^B B_t$$

$$dV_t = \Phi_t^S dS_t + \Phi_t^B dB_t$$

Checking if $V_t$ is a martingale

$$
\begin{aligned}
d\left(\frac{V_t}{B_t}\right) &= dV_t * \frac{1}{B_t} + V_t * d\left(\frac{1}{B_t}\right) \\
&= \Phi_t^S dS_t + \Phi_t^B dB_t * \frac{1}{B_t} + V_t + V_t = \Phi_t^S S_t + \Phi_t^B B_t * d\left(\frac{1}{B_t}\right) \\
&= \Phi_t^s dS_t * \left(\frac{1}{B_t}\right) + \Phi_t^B dB_t * \left(\frac{1}{B_t}\right) + \Phi_t^s S_t * \left(-\frac{1}{B_t}\right) r dt + \Phi_t^B B_t * \left(-\frac{1}{B_t}\right) r dt \\
&= \Phi_t^s \left[-r S_t * \left(\frac{1}{B_t}\right) dt + \left(\frac{1}{B_t}\right) dS_t\right] + \Phi_t^B \left[\left(\frac{1}{B_t}\right) * r B_t dt - r B_t * \left(\frac{1}{B_t}\right) dt\right] \\
&= \Phi_t^s \left[-r S_t * \left(\frac{1}{B_t}\right) dt + \left(\frac{1}{B_t}\right) dS_t\right] \\
&= \Phi_t^s \left[S_t * d\left(\frac{1}{B_t}\right) + \left(\frac{1}{B_t}\right) * dS_t\right] \\
&= \Phi_t^s * d\left(\frac{S_t}{B_t}\right) \\
&= \Phi_t^s dZ_t
\end{aligned}
$$

Under $\mathbb{Q}$ measure

$$
d\left(\frac{V_t}{B_t}\right) = \Phi_t^s dZ_t \quad = \Phi_t^s * \sigma Z_t dW_t^{\mathbb{Q}}
$$

Let

$$
G_t = \frac{V_t}{B_t}
$$

Then $G_t$ is a martingale under $\mathbb{Q}$ measure

Remark the definition of martingale $\forall s < t, \mathbb{E}[X_t|\mathcal{F}_s] = X_s$, $X_t$ is a Martingale.

Therefore

$$
\mathbb{E}^{\mathbb{Q}}[G_T|\mathcal{F}_t] = G_t
$$

$\chi(t, S_t)$ is the option value. At time $T$, the option value is $\chi(T, S_T)$, $S_T$ is the underlying price at time $T$.

$$
\begin{aligned}
V_t &= G_t * B_t \\
&= e^{-rt}\mathbb{E}^{\mathbb{Q}}[G_T|\mathcal{F}_t] \\
&= e^{-rt}\mathbb{E}^{\mathbb{Q}}[V_T * e^{rT}|\mathcal{F}_t] \\
&= e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[V_T|\mathcal{F}_t] \\
&= e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[\max\{S_T - K, 0\}|\mathcal{F}_t]
\end{aligned}
$$

Therefore

$$
V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[\max\{S_T - K, 0\}|\mathcal{F}_t]
$$

## Pricing Call Option

We want to price the value of a call option at $t = 0$

$$
V_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[\max\{S_T - K, 0\}]
$$

Let's introduce an indicator function

$$
V_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)\mathbf{1}_{\{S_T \geq K\}}]
$$

Under $\mathbb{P}$ measure

$$
dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}
$$

Under $\mathbb{Q}$ measure

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

$$S_t = S_0 e^{(r-\frac{1}{2}\sigma^2)t+\sigma\varepsilon\sqrt{t}}$$

and

$$S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)t+\sigma\varepsilon\sqrt{T}}$$

$$\begin{aligned}
V_0 &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)\mathbf{1}_{\{S_T\geq K\}}] \\
&= e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T\geq K\}}] - e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[K\mathbf{1}_{\{S_T\geq K\}}\right]
\end{aligned}$$

**Lenma 1.2.1** Change of Measure Identity

$$\mathbb{E}\left[e^{m+\lambda\varepsilon}\mathbf{1}_{\{\varepsilon\geq a\}}\right] = e^{\frac{\lambda^2}{2}+m}\mathbb{N}(\lambda - a)$$

where $\varepsilon \sim \mathbb{N}(0,1)$, $m, \lambda, a$ are constant.

***Proof:***

$$\begin{aligned}
\mathbb{E}\left[e^{m+\lambda\varepsilon}\mathbf{1}_{\{\varepsilon\geq a\}}\right] &= \int_{-\infty}^{\infty} e^{m+\lambda x}\mathbf{1}_{\{x\geq a\}}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx \\
&= \int_{a}^{\infty} e^{m+\lambda x}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx \\
&= \int_{a}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\frac{(x-\lambda)^2}{2}}e^{\frac{\lambda^2}{2}+m}dx \\
&= \int_{a-\lambda}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\frac{y^2}{2}}e^{\frac{\lambda^2}{2}+m}dy \\
&= e^{\frac{\lambda^2}{2}+m}[\mathcal{N}(\infty) - \mathcal{N}(a-\lambda)] \\
&= e^{\frac{\lambda^2}{2}+m}[1 - \mathcal{N}(a-\lambda)] \\
&= e^{\frac{\lambda^2}{2}+m}\mathcal{N}(\lambda - a)
\end{aligned}$$

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$$\begin{aligned}
e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T\geq K\}}] &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[S_0\exp\left[(r-\frac{1}{2}\sigma^2)T + \sigma\varepsilon\sqrt{T}\right]\mathbf{1}_{\{S_0\exp[(r-\frac{1}{2})T+\sigma\varepsilon\sqrt{T}]\geq K\}}\right] \\
&= e^{-rT}S_0\exp\left[(r-\frac{1}{2}\sigma^2)T\right]\mathbb{E}\left[\exp^{\sigma\varepsilon\sqrt{T}}\mathbf{1}_{\left\{\varepsilon\geq\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right\}}\right] \\
&= S_0\exp\left(-\frac{1}{2}\sigma^2T\right)\mathbb{E}^{\mathbb{Q}}\left[\exp^{0+\sigma\sqrt{T}\varepsilon}\mathbf{1}_{\left\{\varepsilon\geq\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right\}}\right] \\
&= S_0\exp\left(-\frac{1}{2}\sigma^2T\right) * \exp\left(\frac{\sigma^2T}{2}+0\right) * \mathbb{N}\left(\sigma\sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= S_0 * \mathbb{N}\left(\sigma\sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= S_0\mathbb{N}\left(\frac{\ln\frac{S_0}{K} + (r+\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= S_0\mathbb{N}(d_1)
\end{aligned}$$

$$
\begin{aligned}
e^{-rT}\mathbb{E}\left[K\mathbf{1}_{\{S_T\geq K\}}\right] &= Ke^{-rT}\mathbb{E}\left[\mathbf{1}_{\{S_T\geq K\}}\right]\\
&= Ke^{-rT}\left[1*Q\left(S_T\geq K\right)+0*Q(S_t\geq K)\right]\\
&= Ke^{-rT}*Q(S_T\geq K)\\
&= Ke^{-rT}Q\left(\varepsilon\geq\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right)\\
&= Ke^{-rT}\mathbb{N}\left(\frac{\ln(\frac{S_0}{K})+(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right)\\
&= Ke^{-rT}\mathbb{N}(d2)
\end{aligned}
$$

Therefore, the call option price

$$V_0 = S_0\mathbb{N}(d_1)-Ke^{-rT}\mathbb{N}(d2)$$

## 1.3 Delta-Hedging PDE Derivation

Considering we have an option and we want to eliminate the risk by buying some shares of stock.

$$\Pi = V-\Delta S$$

Its differential equation is

$$d\Pi = dV-\Delta dS$$

Remark $dS$ and $dV$

$$
\begin{aligned}
dS_t &= \mu S_t dt+\sigma S_t dW_t\\
dV_t &= \left\{\frac{\partial V}{\partial t}+\frac{1}{2}\sigma^2S^2\frac{\partial^2V}{\partial S^2}\right\}dt+\frac{\partial V}{\partial S}dS
\end{aligned}
$$

Pluging $dV_t$ to $d\Pi$, we have

$$d\Pi_t = \left\{\frac{\partial V}{\partial t}+\frac{1}{2}\sigma^2S^2\frac{\partial^2V}{\partial S^2}\right\}dt+\left(\frac{\partial V}{\partial S}-\Delta\right)dS$$

Since

$$\Delta = \frac{\partial V}{\partial S}$$

Then

$$d\Pi_t = \left\{\frac{\partial V}{\partial t}+\frac{1}{2}\sigma^2S^2\frac{\partial^2V}{\partial S^2}\right\}dt$$

Since the hedged portfolio is locally riskless, by no-arbitrage it must earn the risk-free rate. We have

$$d\Pi_t = \Pi r dt$$

Therefore

$$\left\{\frac{\partial V}{\partial t}+\frac{1}{2}\sigma^2S^2\frac{\partial^2V}{\partial S^2}\right\}dt = r\left(V-S\frac{\partial V}{\partial S}\right)dt$$

Removing right side two terms to left side, we get

$$\frac{\partial V}{\partial t}+rS\frac{\partial V}{\partial S}+\frac{1}{2}\sigma^2S^2\frac{\partial^2V}{\partial S^2}-rV = 0$$

**Theorem 1.3.1** Feynman-Kac
Assume that $F$ is a solution to the boundary value problem:

$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$
$$F(T,x) = \Phi(S)$$

Then $F$ has the representation

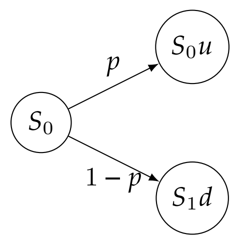
$$F(t,x) = e^{-r(T-t)}\mathbb{E}[\Phi(S(T))]$$

The call option price is

$$V = e^{-rT}\mathbb{E}[(S_T - K)^+]$$

## 1.4 Discrete-Time Delta Hedging: A Binomial Replication Example

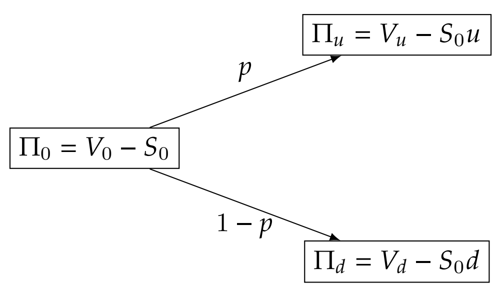
Consider a single period binomial tree




We have a portfolio with a option and stock, $V$ and $S$ are the values respectively. $\Pi$ is the portfolio value. We want to use delta hedging to elimiate the risk, which is shown by

$$\Pi = V - \Delta S$$

where $\Delta$ is the number of the shares.




We want to pick a $\Delta$ to make the equation holding

$$\Pi_u = \Pi_d, \quad V_u - S_0u = V_d - S_0d$$

The $\Delta$ is solved by

$$\Delta = \frac{V_u - V_d}{S_0(u-d)}$$

Assuming maturity is $\delta t$ and the discounted factor

$$D = \frac{1}{B} = e^{-r\delta t}$$

Under assumption of arbitrage free,

$$\Pi_0 = e^{-r\delta t}[V_u - \Delta S_u] = e^{-r\delta t}[V_d - \Delta S_d]$$

Namely

$$V_0 - \Delta S_0 = e^{-r\delta t}[V_u - \Delta S_0 u] = e^{-r\delta t}[V_d - \Delta S_0 d]$$

$V_0$ is solved as

$$V_0 = \frac{V_u(1 - de^{-r\delta t}) - V_d(ue^{-r\delta t} - 1)}{u - d}$$

In risk neutral world, investors must earn the risk-free rate on average. We have

$$\begin{aligned} S_0 e^{r\delta t} &= \mathbb{E}[S_{\delta t}] \\ &= q * S_0 u + (1 - q) S_0 d \end{aligned}$$

The probability of moving up under $\mathbb{Q}$ is

$$q = \frac{e^{\delta t} - d}{u - d}$$

The option price $V$ value at $t = 0$ is

$$V_0 = e^{-r\delta t}[q * V_u + (1 - q) * V_d]$$

Then we need to find a martingale $\mathbb{Q}$ measure for which the discounted stock price is a martingale. Under the $\mathbb{Q}$ measure, the probability of moving up is $q$. Since martingale equation, we have

$$\mathbb{E}^{\mathbb{Q}}[e^{-r\delta t} S_T] = S_0$$

$$e^{-r\delta t}[q * S_0 u + (1 - q) * S_0 d] = S_0$$

We solve $q$

$$q = \frac{e^{r\delta t} - d}{u - d}$$

**Theorem 1.4.1** Fundamental Asset Pricing Theorem

A financial market with time horizon $T$ and price processes of the risky asset and riskless bond given by $S_1, \ldots, S_T$ and $S_1^0, \ldots, S_T^0$, respectively, is arbitrage-free under the probability $\mathbb{Q}$ if and only if there exists another probability measure $\mathbb{Q}$ such that

i For any event $A$, $P(A) = 0$ if and only if $Q(A) = 0$. We say in this case that $\mathbb{P}$ and $\mathbb{Q}$ are equivalent probability measures.

ii The discounted price process, $X_0 := S_0/S_0^0, \ldots, X_T := S_T/S_T^0$ is a martingale under $\mathbb{Q}$.

Therefore $\mathbb{P}$ and $\mathbb{Q}$ are equivalent probability measures.
Therefore, the option prive at $t = 0$ can be written as

$$V_0 = \mathbb{E}^{\mathbb{Q}}[e^{-r\delta t} V_T] = e^{-r\delta t}[q * V_u + (1 - q) V_d]$$

Since Cox-Ross-Rubinstein, we have

$$u = e^{\sigma\sqrt{\delta t}}, \quad d = \frac{1}{u}, \quad D = e^{-r\delta t}$$

$$V_0 = \frac{V_u(1 - de^{-r\delta t}) - V_d(ue^{-r\delta t} - 1)}{u - d}$$

Multiply by $(u - d) * e^{r\delta t}$ both sides

$$(u - d)V_0 * e^{r\delta t} = e^{r\delta t}(V_u - V_d) + (uV_d - dV_u)$$

Using Taylor Expansion, we have

$$\begin{aligned} u &= e^{\sigma\sqrt{\delta t}} \approx 1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2 \delta t \\ d &= \frac{1}{u} \approx 1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2 \delta t \\ e^{r\delta t} &\approx 1 + r\delta t \end{aligned}$$

## 1.5 Self-Financing Replication Derivation

We set the $\chi$ is a option, and $G(S_t)$ is the value at the expiry date.
$V_t$ is the value of the option at the expiry date.

$$V_t = V_0 + \int_0^t \Phi_u^S \mathrm{d}S_u + \Phi_t^B \mathrm{d}B_u \tag{1.5}$$

**Note:-**

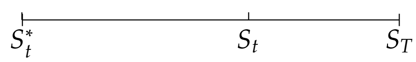
$$\mathrm{d}S_t = \mu S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t$$

To derivative Black-Scholes, we have to solutions that kills $\mathrm{d}W_t$ or $\mathrm{d}t$.

1. kill $\mathrm{d}W_t$ and remain $\mathrm{d}t$, hedge risk (uncertainty) long call and short $\Delta$ stock (risk management) $\longrightarrow PDE$
2. kill $\mathrm{d}t$ and remain $\mathrm{d}W_t$, we get $f(t, S_t)\mathrm{d}W_t$, which is a martingale

Set

$$S_t^* = S_t e^{-rt}$$




$$\mathrm{d}(X_t, Y_t) = X_t \mathrm{d}Y_t + Y_t dX_t + \mathrm{d}X_t \mathrm{d}Y_t$$
$$\mathrm{d}e^{-rt} = -re^{-rt}\mathrm{d}t$$
$$\mathrm{d}t^2 = 0$$
$$\mathrm{d}t\mathrm{d}W_t = 0$$
$$\mathrm{d}W_t^2 = dt$$

we have

$$\begin{aligned}
\mathrm{d}S_t^* &= \mathrm{d}(S_t e^{-rt}) \\
&= S_t \mathrm{d}e^{-rt} + e^{-rt}\mathrm{d}S_t + \mathrm{d}S_t \mathrm{d}e^{-rt} \\
&= S_t(-e^{-rt}\mathrm{d}t) + e^{-rt}(\mu S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t) \\
&= (\mu - r)S_t^* \mathrm{d}t + \sigma S_t^* \mathrm{d}W_t
\end{aligned}$$

In general, $\mu - r > 0$ under $\mathbb{P}$ measurement. We can find $\mathbb{Q}$ st. $\left\{S_t^*, 0 \le t \le T\right\}$ is a $\mathbb{Q}$-Martingale, where $\mu - r = 0$.
We need to prove $\mathrm{d}S_t^* = \sigma S_t^* \mathrm{d}W_t$
The $\theta_s$ is unknown, we need to solve it.

$$\begin{aligned}
\mathrm{d}S_t^* &= (\mu - r)S_t^* \mathrm{d}t + \sigma S_t^* \mathrm{d}W_t \\
&= (\mu - r)S_t^* \mathrm{d}t + \sigma S_t^*(\mathrm{d}W_t^{\mathbb{Q}} - \theta_t \mathrm{d}t) \\
&= \mu S_t^* \mathrm{d}t - rS_t^* \mathrm{d}t + \sigma S_t^* \mathrm{d}W_t^{\mathbb{Q}} - \sigma S_t^* \theta_t \mathrm{d}t \\
&= (\mu - r - \sigma\theta_t)S_t^* \mathrm{d}t + \sigma S_t^* \mathrm{d}W_t^{\mathbb{Q}}
\end{aligned}$$

Let $\mu - r - \theta_t = 0$ to kill drift $(\mathrm{d}t)$, under $\boxed{\theta_t = \dfrac{(\mu - r)}{\sigma}}$ (Sharpe Ratio)

Finally, we get

$$\mathrm{d}S_t^* = \sigma S_t^* \mathrm{d}W_t^{\mathbb{Q}}$$

and we conclude that $S_t^*$ is a $\mathbb{Q}$-Martingale.

**Definition 1.5.1: Martingale**

$\forall s < t \implies \mathbb{E}[X_t|\mathcal{F}_s] = X_s$, $X_t$ is a Martingale.

$\chi(t, S_t)$ is the option value and $V_t$ is the value of the portfolio replicated by risk-free asset and stocks. So, $\chi(t, S_t) = V_t$. We set $V_t^*$ is the discounted value of $V_t$, where $V_t^* = V_t e^{-rt}$.

$$\begin{aligned}
\mathrm{d}V_t^* &= \mathrm{d}(V_t e^{-rt}) \\
&= V_t \mathrm{d}e^{-rt} + e^{-rt}\mathrm{d}V_t + \mathrm{d}V_t \mathrm{d}e^{-rt} \\
&= (\Phi_t^S S_t + \Phi_t^B B_t) r e^{-rt}\mathrm{d}t + (\Phi_t^S S_t + \Phi_t^B B_t) e^{-rt} \\
&= \Phi_t^s(-rS_t + e^{-rt}\mathrm{d}t + e^{-rt}\mathrm{d}S_t) \\
&= \Phi_t^s(S_t \mathrm{d}e^{-rt} + r^{-rt}\mathrm{d}S_t) \\
&= \Phi_t^s \mathrm{d}(S_t e^{-rt}) \\
&= \Phi_t^s \mathrm{d}S_t^*
\end{aligned}$$

$$\mathrm{d}V_t^* = \Phi_t^s \mathrm{d}S_t^*$$

Given $\mathrm{d}S_t^* = \sigma S_t^* \mathrm{d}W_t^{\mathbb{Q}}$
We get

$$\mathrm{d}V_t^* = \Phi_t^s \sigma S_t^* \mathrm{d}W_t^{\mathbb{Q}}$$

$$\mathbb{E}[V_T|\mathcal{F}_t] = V_t^*$$

$$V_T^* = V_T e^{-rt} = \chi(T, S_T)e^{-rt} = G(S_T)e^{-rt}$$

$$\mathbb{E}[V_T^*|\mathcal{F}_t] = \mathbb{E}[G(S_T)e^{-rt}|\mathcal{F}_t]$$

**Claim 1.5.1** Call Option Pricing

$$C = e^{-rt}\mathbb{E}[(S_T - K)\,\mathbf{1}_{\{S_T \geq K\}}]$$

**Lenma 1.5.1**

$$\mathbb{E}\left[e^{m+\lambda\varepsilon}\mathbf{1}_{\{\varepsilon\geq a\}}\right] = e^{\frac{\lambda^2}{2}+m}\mathbb{N}(\lambda - a)$$

where $\varepsilon \sim \mathbb{N}(0,1)$, $m, \lambda, a$ are constant.

***Proof:***

$$\begin{aligned}
\mathbb{E}\left[e^{m+\lambda\varepsilon}\mathbf{1}_{\{\varepsilon\geq a\}}\right] &= \int_{-\infty}^{\infty} e^{m+\lambda x}\mathbf{1}_{\{x\geq a\}}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx \\
&= \int_a^{\infty} e^{m+\lambda x}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx \\
&= \int_a^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\frac{(x-\lambda)^2}{2}}e^{\frac{\lambda^2}{2}+m}dx \\
&= \int_{a-\lambda}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\frac{y^2}{2}}e^{\frac{\lambda^2}{2}+m}dy \\
&= e^{\frac{\lambda^2}{2}+m}[\mathcal{N}(\infty) - \mathcal{N}(a-\lambda)] \\
&= e^{\frac{\lambda^2}{2}+m}[1 - \mathcal{N}(a-\lambda)] \\
&= e^{\frac{\lambda^2}{2}+m}\mathcal{N}(\lambda - a)
\end{aligned}$$

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***Proof:***

$$
\begin{aligned}
e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T\geq K\}}] &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[S_0\exp[(r-\frac{1}{2})T+\sigma\varepsilon\sqrt{T}]\mathbf{1}_{\{S_0\exp[(r-\frac{1}{2})T+\sigma\varepsilon\sqrt{T}]\geq K\}}\right] \\
&= e^{-rT}S_0\exp[(r-\frac{1}{2}\sigma^2)T]\mathbb{E}\left[\exp^{\sigma\varepsilon\sqrt{T}}\mathbf{1}_{\left\{\varepsilon\geq\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right\}}\right] \\
&= S_0\exp\left(-\frac{1}{2}\sigma^2T\right)\mathbb{E}^{\mathbb{Q}}\left[\exp^{0+\sigma\sqrt{T}\varepsilon}\mathbf{1}_{\left\{\varepsilon\geq\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right\}}\right] \\
&= S_0\exp\left(-\frac{1}{2}\sigma^2T\right)*\exp\left(\frac{\sigma^2T}{2}+0\right)*\mathbb{N}\left(\sigma\sqrt{T}-\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= S_0*\mathbb{N}\left(\sigma\sqrt{T}-\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= S_0\mathbb{N}\left(\frac{\ln\frac{S_0}{K}+(r+\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= S_0\mathbb{N}(d_1)
\end{aligned}
$$

$$
\begin{aligned}
e^{-rT}\mathbb{E}\left[K\mathbf{1}_{\{S_T\geq K\}}\right] &= Ke^{-rT}\mathbb{E}\left[\mathbf{1}_{\{S_T\geq K\}}\right] \\
&= Ke^{-rT}\left[1*Q\left(S_T\geq K\right)+0*Q(S_t\geq K)\right] \\
&= Ke^{-rT}*Q(S_T\geq K) \\
&= Ke^{-rT}Q\left(\varepsilon\geq\frac{\ln(\frac{K}{S_0})-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= Ke^{-rT}\mathbb{N}\left(\frac{\ln(\frac{S_0}{K})+(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= Ke^{-rT}\mathbb{N}(d2)
\end{aligned}
$$

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**Claim 1.5.2** Call Option Pricing

$$C = S_0\mathbb{N}(d_1) - Ke^{-rt}\mathbb{N}(d_2)$$

## 1.6 Comparison of the Three Perspectives

The risk-neutral martingale approach derives the option price by changing from the physical probability measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$. Under $\mathbb{Q}$, discounted asset prices become martingales, and the derivative price can be written as the discounted expected payoff.

The delta-hedging approach derives the Black-Scholes PDE by constructing a locally riskless portfolio. By choosing $\Delta$ equal to the derivative of the option value with respect to the stock price, the stochastic risk term is eliminated.

The self-financing replication approach shows that the option payoff can be replicated by dynamically trading the stock and bond. Therefore, by the no-arbitrage principle, the option price must equal the cost of the replicating portfolio.

These three perspectives are closely related. Delta hedging constructs the replicating strategy, replication justifies the no-arbitrage price, and martingale pricing provides the compact expectation formula.

Although the Black-Scholes formula can be derived in many different ways, these derivations are not completely independent. Most of them are different mathematical representations of the same no-arbitrage principle in a complete market. Delta hedging, self-financing replication, and risk-neutral martingale pricing form the three core perspectives. Other methods, such as Feynman–Kac, Girsanov theorem, heat-equation transformation, lognormal integration, change of numeraire, and the binomial limit, can be viewed as refinements or extensions of these core perspectives.